RACE CAR | FULL HARNESS

INDIVIDUAL PHYSICAL CONDUCTORS / REV J / 27 OCT 2026

116 native nets | 179 field conductors | 85 supplier paths | 2 intact cable assemblies | 45 explicit joint / ground nodes

PROPOSED CONNECTION TOPOLOGY - unmeasured lengths, wire sizes, colors, terminals and splice hardware. NOT A FABRICATION RELEASE.

—— C / FIELD PATH
------ S / SUPPLIED LOOM
—— V / VENDOR CABLE
- - - - I / INTERNAL
-·-·- B / BOND / METAL
········ U / UNRESOLVED

BUILD TAGS: MAKE = provide branch | LEAD = supplied wire, finish ends | LOOM = existing path | CABLE = intact cable | HOLD = verify

Black bars inside boxes are explicit splices/common metal/shared terminals. Crossings with white gaps never join.

Find by wire ID, net ID, component, joint or PDF bookmark. Click a component header for the complete native sheet.

VECTOR / SEARCHABLE | 196.1 x 100.2 inches

Line geometry is illustrative; vehicle routes and cut lengths are unmeasured.

02 / BATTERY, ISOLATION AND STARTER
BT1 Battery
G1 Alternator
F0 MAIN BUS FUSE / SELECT
F3 ALT BRANCH FUSE / AT STARTER
F1 Engine controller supply fuse
U3 Cole Hersee 24059 starter relay
D60 Starter-relay coil suppression diode
M1 Starter motor
J1 Engine controller power connector
F5 START DRIVER FUSE / SELECT
F2 Shared starter / charging cable fuse
J2 Engine ground strap
J3 Chassis ground strap
03 / NEXUS R3 PIN ALLOCATION
J10 Engine controller - connector A
J11 Engine controller - connector C
04 / ENGINE POWER, INJECTION AND IGNITION
J12 Haltech R3 - integrated PDM outputs
INJ1 LS3 injector - cylinder 1
OEM111-PWR Stock odd coil PWR splice
OEM111-SG Stock odd coil SG splice
OEM113-PG Stock even coil PG splice
OEM-INJ-ODD Stock odd injector feed splice
OEM-INJ-EVEN Stock even injector feed splice
CUT-INJ1 Donor injector 1 control cut-end
CUT-INJ3 Donor injector 3 control cut-end
CUT-INJ5 Donor injector 5 control cut-end
CUT-INJ7 Donor injector 7 control cut-end
M3 Fuel lift pump 1
M4 Fuel lift pump 2
M5 High-pressure fuel pump
B11 Haltech HT-010910 fuel pressure
M6 Radiator fan 1
M7 Radiator fan 2
B12 Fuel level sender
07 / FUEL PUMPS, SENDERS AND COOLING
J20 Anti-lock brake unit - main connector
08 / DELTA ABS GEN 2 MAIN CONNECTOR
B23 Rear right wheel-speed sensor
B24 Front brake pressure sensor
B25 Rear brake pressure sensor
S20 Anti-lock brake slip selector
J22 Anti-lock brake vehicle interface
S21 Anti-lock brake enable switch
H20 Anti-lock brake warning lamp
S1 Engine controller power switch
U10 Haltech iC-7 digital dash
U11 Driver control keypad
J30 PD16 internal termination / representation
J31 Vehicle data-bus termination
B30 Transaxle gear-position sensor
10 / CAN, DASH, KEYPAD AND WAKE
J41 Haltech PD16 - 25 A power outputs
U20 Haltech PD16 - main power
F50 PD16 main supply fuse
H30 Left LED headlight
H37 Right LED headlight
H31 Left tail LED bulb
H35 Right tail LED bulb
H34 Left brake LED bulb
H36 Right brake LED bulb
U22 Brake-marker electronics
11 / AUXILIARY DISTRIBUTION AND LIGHTING
J40 Haltech PD16 - connector A
13 / HALTECH PD16 CONNECTIONS AND CONTROL
U24 WolfBox G900 Pro hardwire converter
U21 WolfBox G900 Pro mirror and front camera
U25 WolfBox rear camera
17 / WOLFBOX G900 PRO CAMERA POWER
GP009 Engine block / coil ground point
SP026 R3 CAN RETURN splice
SP032 COCKPIT 12V splice
CAN1-1H CAN1 station 1 / H
CAN1-2H CAN1 station 2 / H
CAN1-3H CAN1 station 3 / H
CAN1-1L CAN1 station 1 / L
CAN1-2L CAN1 station 2 / L
CAN1-3L CAN1 station 3 / L
CAN2-ENDH CAN2 far-end tap / H
CAN2-ENDL CAN2 far-end tap / L
SP061 R3 REF5 splice
SP062 R3 SG splice
SP089 ABS PRESS 5V splice
SP103 ABS WAKE splice
SP117 TAIL POS splice
SP121 CAMERA POS splice
SP136 GT blue / blue permanent joint
OEM111-PG Stock odd coil PG splice
OEM113-PWR Stock even coil PWR splice
OEM113-SG Stock even coil SG splice
CUT-INJ-ODD-PWR Donor odd injector feed cut-end
CUT-INJ-EVEN-PWR Donor even injector feed cut-end
CUT-INJ2 Donor injector 2 control cut-end
CUT-INJ4 Donor injector 4 control cut-end
CUT-INJ6 Donor injector 6 control cut-end
CUT-INJ8 Donor injector 8 control cut-end
EXPLICIT SPLICES / PROPOSED GROUND POINTS / SUPPLIER JOINS

PHYSICAL CONNECTION NOTES / NATIVE DESIGN HOLDS

All new joint/ground locations and material choices remain proposed. Electrical endpoint sets are checked against the native KiCad export.

PHYSICAL CONDUCTOR VIEW - READ FIRST

WHAT YOU BUILD: C / MAKE = connection you provide, using new wire or suitable reused donor wire/pigtail. S / LOOM = existing OEM/vendor loom wire; retain it. V / CABLE = intact supplied assembly to connect.

Read MAKE / LEAD / LOOM / CABLE tags, not ID prefixes, for construction work. LEAD = supplied harness wire needing end work; MAKE = additional connection/branch to provide. Inspect actual grouped leads before adding wire.

R3 HT-183200 and PD16 HT-188000 flying-lead harnesses are owner selected. Controller-side leads are supplied; free ends need routing/termination. Heavy main feeders and vehicle branches are separate. A drawn LEAD span can include an extension or adapter.

Delta loom, sensor branches, two pressure plugs, slip/OFF switches, warning lamp, fused feeds and IMU/private CAN are supplied assemblies. Do not fabricate duplicate branches or add sensor grounds. IMU cavities remain supplier-internal/unmapped.

Delta switch HOLD: current guide describes hardware solenoid-power interruption; historical pin35 ground-enable model is unresolved. C158 is HOLD, not a wire to make. Keep actual supplied switch loom intact pending revision verification.

Delta F41/F42/F43 represent supplied loom protection: current guide lists pump40A/solenoid20A/ignition10A; older pinout labels a25A supply. Confirm received loom before final coordination. Public CAN uses auxiliary port; private four-pin port is service only.

I / INTERNAL and B / BOND are not extra cut wires. U / HOLD is unresolved supplier construction. Wires_To_Provide.tsv and Supplied_Leads_To_Finish.tsv separate added wiring from lead end work; neither is a measured cut schedule.

Each solid C line is one individual proposed field conductor with exactly two ends. Bends and white crossing breaks do not add wire ends. W identifiers still name electrical nets.

S lines are individual supplied-loom conductor paths, retained within their supplier assembly; verify actual construction. V thick purple lines are intact multi-wire vendor cables, not single cut wires.

Short black bars inside SP nodes denote explicit proposed splices; inside TB1/TB2 they denote conducting metal. Repeated same pin labels share one actual terminal/stud; they are not additional cavities.

U3 is Cole Hersee 24059: large PWR from F5/2, large OUT to M1 small SOL; small CMD from R3 A18/HBO5 and small GND to GP129-C. These are installer tags, not stamped terminal numbers. Coil and contact supplies are separate.

D60 1N5408 is local across U3 small coil studs: band/K to CMD via C213; unbanded/A to GND via C214. C213/C214 are insulated diode leads, not long harness cuts. The W129 ground-stub depiction of C214 does not require another chassis landing. Verify polarity and prompt disengagement.

S2 clutch/neutral switch is removed by owner; C076/C123 and W046 retired. R3 C5/SPI1 now serves S50 HT-010504 TC trim, never jumpered. Actual start logic must omit that input requirement while retaining fresh request, engine-not-running, bounded crank and release controls.

Dashed I lines denote internal device/supplier continuity; dash-dot B lines denote metal/lug/bond continuity. Neither is an extra cut wire. Dotted U lines remain unresolved supplier paths.

Splice and ground-point positions, hardware, joining method, wire gauge/color, route, cut length and terminal parts are UNRELEASED. This diagram proposes connection topology, not measured vehicle geometry.

Ground points GP129-F/C/R are proposed front/control/rear vehicle-side returns. Their metal/bond network must be verified; no extra field wire is specified by dash-dot structure paths.

Chassis returns use short stubs and labelled ground symbols. Match W IDs; the graphic stub is not a measured cut length. GP129-F/C/R remain distinct proposed physical landing points on W129.

B005 GP129-F to GP129-C and B006 GP129-C to GP129-R are proposed chassis metal/bond continuity, represented by the W129 ground references, not extra drawn field wires.

W015 chassis/case bond references stay distinct from W129 return/strap terminals. No ground symbol is substituted for W002 isolated battery negative, W009 engine returns, sensor grounds, CAN returns or supplied Delta sensor returns.

J2/J3 ground straps and J21 ground assembly have two ends joined by their own assembly bodies. Those bodies are depicted inside their boxes, not as additional field wires.

CAN1 uses a proposed linear trunk R3 -> dash tap -> keypad tap -> TMS4 tap -> PD16. H/L pair IDs match each trunk/tap. J30 is the PD16 end-termination representation; no third resistor is added.

TB1 four branch wires and TB2 six branch wires (incoming directly from F0) land at explicit functional lug ports. Actual stud/stack assignments and block ratings/boots remain open. M1 B+ has two wires on the same battery-side stud.

Existing device pin/net assignments are retained. Stock L94 X111/X113 bank connector reference interfaces are added; functional device IDs do not imply inspected donor cavities.

S paths marked OEM-COIL or OEM-INJ are retained donor-loom conductors, not new branch wires. OEM splice nodes depict shared continuity; exact internal locations, wire sizes and condition require inspection.

Stock injector integration uses functional CUT-INJ boundaries, not an invented factory bulkhead plug. Isolate injector-only feed branches from factory coil/other loads; never bridge HCO1 and HCO2.

Stock coil low-reference grounding remains HOLD: retained C26 routing is not final approval; resolve GM separate low reference versus Haltech generic head-ground advice with actual coil/R3 evidence.

Use Physical_Connections.tsv for ID/from/to/kind/pair and blank measurement fields, Physical_Junctions.tsv for joint intent, and Physical_Topology.json as the maintained topology source. Never overwrite measured entries by re-initializing it.

09 / ABS SENSORS, CONTROLS AND PUBLIC CAN

Wheel sensors -> Delta ABS -> public CAN -> R3 traction control. Selected path; Gen 2 protocol check remains open.

HOLD A03: GM 22742183 signal compatibility/polarity unverified. Check both corners; GM lists it as front-left assembly.

Both pressure sensors use labelled 3-pin Delta loom plugs. Return W112 is supplier-internal; no added ground wire or R3 tap.

IMU/yaw sensor and its private CAN remain in supplied harness: map with verified harness data before build release.

S21/C158 HOLD: current guide describes hardware solenoid-power OFF; old pin35 ground-enable map is not a fabrication instruction.

CAN wheel speeds selected per Haltech support. J22:7-10 unused; verify Gen 2 profile/bitrate (sheet 15). SPI2 = brake request.

13 / HALTECH PD16 CONNECTIONS AND CONTROL

Owner uses HT-188000 flying-lead harness | PD16 diagram rev 1.5, October 2024 | Peripheral Power Mode / Voltage Sense

ONE MODULE - THREE DRAWING REFERENCES

U20 = main positive; J41 = connector B; J40 = connector A.

Peripheral mode: dedicated R3 HBO6/A25 -> PD16 A25.

A13 and A26 are unused. Do not backfeed A25 as an output.

Ground A31-A34 to VEHICLE_RETURN, not BT1 negative.

Use four 18 AWG ground wires per Haltech guide.

A22 is CAN cable ground; not a substitute for A31-A34.

FAN CONFIGURATION - NO EXTERNAL PWM

Assign PD16 25A HCO1 / HCO2 to separate thermofans.

Use coolant-temperature on/off hysteresis and keypad override.

Temperature thresholds must match the thermostat/cooling test.

Verify power-up, CAN loss, sensor-fault and overcurrent behavior.

Do not make fan-off on track unconditional; retain overheat demand.

Measure hot-idle charging with both fans on; listed pair load 20 A.

CAN hub branch: H/L/return only. Keep PD16 A25 power separate from cockpit +12V; no four-wire power bridge.

Use the manufacturer connector-face view for cavity orientation. A/B prefixes identify connectors, not additional modules.

A3-A10/A17-A18 outputs are limited to 8 A each; B1-B4 to 25 A each. Main connector rating is not an aggregate thermal guarantee.

Configure Peripheral Power Mode, Haltech CAN 1 Mbit/s and PD16 end termination in NSP. Do not add another resistor at J30.

PD16 manufacturer baseline: main 4 AWG, B outputs 10 AWG, A high-side outputs 18 AWG. Final routing/derating/protection remain open.

HT-188000 supplies controller-side leads and CAN plug. Finish free ends; no cockpit-positive bridge into dedicated PD16 wake.

16 / TMS-4 TPMS, ENGINE AND TRANSMISSION OIL TEMP

Two HT-010306 high-range 150C sensors use separate PD16 AVIs. HT-011600 tyre sensors communicate wirelessly.

Engine: PD16 A20 / AVI1 -> B32:B; A11 -> B32:A.

Transmission: PD16 A21 / AVI2 -> B33:B; A11 -> B33:A.

Enable each AVI internal 1k pull-up; no external 5V wire.

Both: load Haltech Oil 150C 1/8 NPT HT010306.cal.

Separate engine/trans channels; verify curve and sharing.

EV1 harness view: A return / B signal. A11: sensors only.

Supplied DTM4 -> MCON6 adapter:

DTM1 +12V -> MCON5; DTM2 GND -> MCON6.

DTM3 CAN H -> MCON2; DTM4 CAN L -> MCON1.

Receiver: 12-32V; 50mA ambient operating current.

Use supplied adapter; do not infer plug viewing direction.

FOUR WHEEL SENSORS / WIRELESS

Fit internal sensors to FL / FR / RL / RR; batteries are inside the sensors. No wires run out to the wheels.

NSP: Connections > CAN > Haltech CAN System > Tyre Monitoring / TMS-4 Enable; assign IDs to actual corners.

TMS-4: verify corner identity, pressure/temperature units, alarm thresholds and lost-sensor response. Mount receiver high in vehicle.

Manufacturer typical in-motion refresh: 60 seconds. TPMS is for tyre monitoring; it is NOT a wheel-speed source for traction control.

PD16 engine/trans temp: separate open/short diagnostics, valid range, alarms and stale-CAN behavior. Thresholds remain unset.

Confirm trans mounting/immersion, thread clearance, 150C range and calibration against a reference; no mechanical location specified.

Sources: Haltech TMS-4 QSG; HT-010306 info sheet rev 4.2; PD16 diagram rev 1.5. See Sources/ and README.

19 / STOCK L94 COIL LOOMS AND LS3 INJECTORS

Retained donor wiring / 7 populated bank ways / related-year GM cavity reference; confirm actual donor continuity and orientation.

OEM COIL BANK LOOMS - RETAIN THEIR INTERNAL SPLICES

Each bank: four triggers plus common +12V, power ground and low reference. D is unavailable, not an eighth live wire.

Reference coil plug: A=power ground, B=low reference, C=trigger, D=+12V. Device IDs on sheet 04 stay functional.

S paths in the continuous views are RETAINED STOCK wiring. C paths are new integration wires; no new coil-branch cuts assumed.

DONOR INJECTOR WIRING - REUSE WITH LS3 EV6 / USCAR INJECTORS

Retain eight individual driver legs and separate odd/even +12V branches. No factory injector bulkhead plug is assumed.

CUT-INJ labels mark functional donor cut/termination boundaries, NOT connector cavities; locations/join hardware TBD.

Factory feed circuits may also serve coils: isolate retained injector feeds before HCO1. NEVER connect HCO1 to HCO2.

HOLD: verify LS3 injector PN, plug key/latch, connector orientation, harness reach, condition, continuity and calibration.

GROUND HOLD: bank A -> engine; bank E -> R3 C26 is retained pending exact coil evidence. No W009/W062 bridge permitted.

GM separates low reference; Haltech generic Gen IV guide also advises both coil grounds at the head. Resolve with exact coil/R3 guidance.

Reference: Stock_Engine_Harnesses.json and OEM_Harness_Interfaces.tsv. Source cavities/colors are not inspected donor measurements.

No downstream bank fuses restored. HCO1 injector / HCO2 coil trip coordination with every OEM branch and contact remains OPEN.

System index

DANNY'S RACE CAR

ELECTRICAL HARNESS / REV J

L94 / LS3 | Nexus R3 | Delta ABS Gen 2 | iC-7 | 15-button CAN keypad

DESIGN REVIEW DRAFT - OPEN THIS ROOT FILE IN KICAD

WHAT IS VERIFIED

R3 and PD16 connector functions; Delta supplied pin allocations; owner equipment choices. Circuit assignments are a draft.

WHAT REMAINS OPEN

TBD devices, selected ratings, sensor cavity mapping, ABS compatibility, isolation, CAN configuration and physical harness routing.

05 / ENGINE SENSORS AND WIDEBAND

R3 analog allocation is on sheet 03. Sensor model, reference voltage and calibration must match before connection.

MAP SENSOR IS INSIDE NEXUS R3

Onboard 4-bar absolute MAP selected in NSP.

Intake manifold -> pressure hose -> R3 MAP barb.

Pneumatic connection only; no external sensor wires.

AVI9 / C18 spare (onboard MAP). One wideband only.

Crank A -> C1; B -> C26; C -> C25. Cam A -> C25; B -> C26; C -> C3. W078/W079 retired; never reuse.

Shield crank/cam/knock cables; shield/drain to R3 A10 only at ECU end. Sensor return remains C26.

Owner confirms crank GM12703627 and cam GM12591720. Verify molded cavities; C25 200mA includes BOTH sensors.

WBI1 C29-C34: HTR+ / INPUT / PUMP / NERNST / HTR- / CAL. Retain matching sensor/adapter; no separate power/ground.

HT-010206: C17, 1k pull-up, GM Air calibration. O2: select LSU4.9 or NTK in NSP; exact pack/cavities and exhaust location TBD.

08 / DELTA ABS GEN 2 MAIN CONNECTOR

Bosch 1928405167 / 38 cavities | Source: owner-supplied SCS Delta Gen 2 Rev.3 allocation.

Owner retains complete Delta loom and its fuses. Guide: pump40A/solenoid20A/ignition10A; older25A label requires revision check.

J21 records the proposed battery return; HOLD A01: reconcile manual chassis-ground wording with pinout battery-negative wording.

HOLD A02: identify which HCU CAN pair serves public CAN and which serves the supplied IMU/diagnostic harness.

ABS sensor grounds remain within ABS. Main connector unused output pins have NC marks; public-interface detail follows.

Finished Delta branches and IMU/private CAN are supplied, not wires to fabricate. Vehicle power/public-CAN interface only.

11 / AUXILIARY DISTRIBUTION AND LIGHTING

One Haltech PD16 / HT-198100 kit. U20, J40 (sheet 13) and J41 are interfaces on the SAME module.

Control, CAN and all four ground wires: J40 on sheet 13.

Main +12V: 4 AWG manufacturer baseline; F50 rating TBD.

WOLFBOX CAMERA SYSTEM - SHEET 17

A7 / HCO5 -> kit -> USB-C mirror power.

Keypad camera toggle; proposed position 10.

HEADLIGHT_POS = both headlights / PD16 B3. TAIL_POS = both tail bulbs / A3. BRAKE_POS = both brake bulbs / A4.

Keypad hazards: flash headlights AND tail bulbs together. Hazard logic overrides steady ON so the lamps can turn OFF each cycle.

Brake bulbs stay steady on brake request; no hazard flashing of brake bulbs. A5/HCO3 and A6/HCO4 are now SPARE.

Hazards OFF: restore normal head/tail requests. Implement shared flash phase and verify LED-driver cycling; no lamp models selected.

HOLD L01: configure/test flash logic, CAN-loss behavior, bulb currents, conductor/contact protection and repeated-start allowance.

14 / HGT SHIFT KNOB AND FLAT-SHIFT CONTROL

B31 senses driver shift force. B30 (sheet 10) senses actual gear. Both signals are read directly by the Nexus R3.

R3 CONNECTIONS - CONNECTOR C / J11, SHEET 03

Knob cavity 1 -> C26 / sensor signal ground.

Knob cavity 2 -> C28 / AVI11; pull-up disabled.

Knob cavity 3 -> C25 / regulated sensor +5 V.

Internal amplifier: no external amplifier or cut relay.

Shared 5 V load budget: 200 mA maximum at C25.

HGT current demand is unlisted; verify before release.

HARNESS AND INSTALLATION

Proposed harness extension: three 22 AWG automotive wires; verify contact size and routing.

Use the supplied DT-style cavity numbers; verify the actual mating shell and terminal view.

Wire colors are unverified. Do not assign functions by color. Ground returns to C26 only.

Retain lead strain relief and a flexible service loop through full lever travel.

Mount for fore/aft force. M12 x 1.75 fitment; check adapter against the actual Weddle lever.

NSP SETUP - REQUIRED BEFORE ENABLING SHIFT CUT

Select Load Cell as flat-shift trigger; assign AVI11. Keep cut/blip disabled during initial checks.

At the selected 5 V supply, HGT resting output is about 1.80 V. Measure rest, pull and push.

Confirm actual signal direction; calibrate trigger voltages and hysteresis for driver force.

Use B30 / AVI10 calibrated gear feedback for closed-loop shift setup; validate every gear.

Tune RPM/TPS enable limits, cut/recovery, maximum cut time and retrigger blockout.

Verify sensor-disconnect behavior. Auto-blip remains disabled until DBW/downshift tuning.

Source: HGT installation guide 63800-IB-SCK-008, 12 July 2024, especially page 5; see README for links.

Pinout applies to HGT pre-terminated DT-style units. Confirm supplied hardware before terminating the mating harness.

HOLD S01: supply-current budget, mating connector, installed signal calibration and tuner validation. No NSP file programmed.

18 / CARTEK GT, STOP SWITCH AND POSITIVE BLOCKS

GT manual rev1.0 / 07Nov2025, method 1 ECU/PDM input. ONE internal stop switch; no external switch. Blocks are proposed parts.

S40: one cockpit stop button, reachable inside/outside.

Reset/released = NC closed / ON; pressed = latched open / OFF.

Verify actual NC pair. Pins 1/2 are functional, NOT cavity IDs.

Join BLUE_A to BLUE_B permanently; no chassis connection. S41 removed. No battery voltage on switch leads.

GT M6_IGN -> S1 -> R3 A13 only. K1 removed; F0 -> TB2 direct. GT output is fixed 40A protected.

TB1: four lugs / common M10 post, stack and covers TBD. TB2: six lugs / four 3/8 studs; allocation TBD.

TB2 listing conflicts: specific 250A vs family 150A. Confirm actual rating, thermal duty and permitted lug stacks.

GT: 7-18V; 600A cranking / 2000A surge. Confirm received GT-s/GT-x, terminal labels and actual starter current.

Install in manual order: chassis bond, switch/blue loop, engine-stop, M6 battery positive, then M8 negative. Limit temp to +85C pending source conflict.

F4 5A remains PROPOSED for the GT/control feed. Size wiring/protection for actual GT + R3 A13 load, not a contactor coil.

K1 deleted by owner. Verify R3 A13 OFF stops engine/fuel and PD16 outputs promptly before GT battery isolation.

Twist reset may re-arm power; require a fresh start command. Charger/booster cables both at battery posts; no negative bypass.

02 / BATTERY, ISOLATION AND STARTER

CARTEK GT is the master isolator; K1 removed. F0 feeds TB2 directly. Returns are downstream of isolation.

PHYSICAL ROUTE: battery-side BATT_RAW block -> F2 -> M1 battery B+ stud -> F3 -> G1 B+.

U3 / 24059 BULKHEAD RELAY WIRING

Large PWR <- F5/2; large OUT -> M1 SOL.

Small CMD <- R3 A18/HBO5 (+12V ON).

Small GND -> VEHICLE_RETURN (not case).

D60 band/K -> CMD; unbanded/A -> GND.

PWR/OUT/CMD/GND are installer tags.

12V coil: 17.5 ohm / ~0.69A calculated.

U2 isolates NEGATIVE. Positive starter/alternator cables remain battery-connected when OFF: insulate, route and protect them.

GT M6_IGN -> S1 -> R3 A13 is the engine-stop control. F0 -> TB2 is permanently battery-positive when connected.

F5 protects the relay-contact/solenoid branch; value TBD. Estimated 30A solenoid pull-in is NOT a measurement. No motor current through U3.

K1 omitted by owner. Verify R3 ignition-OFF stops injection/ignition/fuel and PD16 loads, including CAN-loss/shutdown timing.

No wire except U2 M8 may return to BT1 negative. R3/PD16 and every load use VEHICLE_RETURN; verify with all service cables.

W011: one shared start/charge trunk. F3 branches at M1 BATTERY B+ stud, never SOL or motor-side stud. Size/protection TBD.

06 / DRIVE-BY-WIRE

Pedal GM 19417903 | Stock LS3 throttle body | Function labels below are NOT physical cavity numbers.

S2 REMOVED / NO CLUTCH SWITCH

R3 C5 / SPI1 now serves S50 TC trim; W046 remains retired.

Fresh start request, engine-not-running and bounded crank remain required.

No clutch/neutral switch input; physical master stop remains independent.

HOLD D01: verify all six pedal cavities and all throttle-body cavities, including motor polarity, before terminating.

Pedal pigtail missing: donor connector may be reusable; CPP-LSAPP is a listed candidate. Housing fit does not prove wiring.

APP1/APP2 -> C10/C11; TPS1/TPS2 -> C12/C13. No pull-ups assumed for powered position signals.

Complete Haltech DBW calibration, track correlation, brake override and fault checks before any powered vehicle test.

Owner removes S2 and its two wires. No clutch/neutral start interlock is drawn; SPI1 must not be jumpered.

10 / CAN, DASH, KEYPAD AND WAKE

CAN1: R3 / PD16 / iC-7 / keypad / TMS-4 (1 Mbit/s Haltech). CAN2: Delta ABS public bus, separately verified.

GEAR DISPLAY - R3 TO IC-7 OVER CAN1

NSP: Gear Position Sensor; AVI10 / C27; pull-up OFF.

Calibrate every forward gear, neutral and reverse in place.

Set non-overlapping voltage windows; check repeatability.

IC-7: display ECU Gear channel. No separate display needed.

KEYPAD / CAMERA CONFIRMED; POSITION 10 PROPOSED

1 Run/Stop 2 Start (momentary) 3 Headlights 4 Tail lights 5 Hazards

6 Fan override 7 Timed fuel prime 8 Dash next 9 Alarm acknowledge 10 Camera ON/OFF

11 Spare 12 Spare 13 Spare 14 Spare 15 Spare

S1 is normal wake, supplied only by CARTEK U2 M6_IGN. Emergency stop uses U2/S40 and R3 A13; verify engine/output shutdown and no backfeed.

CAN termination is a topology requirement, not two extra resistors to fit blindly. R3 has selectable internal termination per bus.

CAN1: R3 termination at one end; choose PD16 at the other and enable its termination. J30 represents that same resistor.

Twist each H/L pair; minimize stubs. Do not tie CAN1 to CAN2. R3 A11 total device return current must stay within its rating.

iC-7 fuel level: AVI4 pin 21 / return 30; 240R pull-up for resistive sender. Confirm dash hardware and NSP firmware.

15 / TRACTION CONTROL AND SPARE I/O

Selected: Delta CAN wheel speeds to R3. Haltech support response supplied by owner; see README / Sources.

SELECTED WHEEL-SPEED PATH

Four wheel sensors -> Delta ABS (sheets 08-09).

Delta public J22:3 CAN H / J22:2 CAN L -> R3 CAN2.

R3 J11/C21 = CAN2 H; J11/C22 = CAN2 L (sheet 03).

NSP: select the available Delta ABS vehicle-CAN profile.

Assign FL, FR, RL and RR speed sources to CAN Input.

R3 compares calibrated front/rear speeds for traction control.

J22:7-10 pulse outputs stay NC; R3 SPI3-6 stay spare.

Sensor signal/return pairs remain connected to Delta only.

SUPPORT CONFIRMATION AND CHECKS

Haltech support: CAN is suitable; Delta profile is available.

Gen 2 qualification: verify protocol against installed NSP.

Confirm public-bus bitrate, termination and harness mapping.

Published wheel-speed broadcast: 100Hz / nominal 10ms.

Check wheel identity, units and calibration for all four wheels.

Validate TC enable, slip tuning and sensor/CAN fault response.

Do not infer receive/TC rate from ECU-to-dash broadcasts.

This drawing does not configure NSP or validate live data.

SPARE CHANNELS - REVISION J

R3: 1 dedicated AVI; 4 SPI; 6 DPO; 0 HBO; 0 HCO.

R3 AVI9/C18 spare (onboard MAP); SPI3-6 remain spare.

Fuel pressure AVI6/C15; intake air temperature AVI8/C17.

PD16: 2 AVI; 4 SPI; 2 HBO; 6 x 8A HCO; 0 x 25A HCO.

PD16 AVI1/A20 engine oil; AVI2/A21 trans; A11 return.

iC-7: 3 spare legacy sensor AVI1-3; AVI4 = fuel level.

NSP additionally exposes AVI5-9, 3 DPI and 4 DPO.

Extended dash capacity is firmware-dependent; see README.

S50:1 -> C25 5V; S50:2 -> C5/SPI1; S50:3 -> C26.

SPI1 analog rotary input; input pull-up MUST be OFF.

Use HT-010504 calibration matching the fitted decal.

TC levels/slip/cut/fault behavior remain tuner setup.

No wheel-sensor split; Delta CAN2 speed path unchanged.

NSP: PD16 AVI1 = Engine Oil Temperature; AVI2 = Transmission Oil Temperature. Separate HT-010306 calibrations.

iC-7 fuel level: pin 21, sensor return pin 30. NSP sharing minimums: R3 1.26.4 / dash 2.26.3; verify installed versions.

TMS-4 uses CAN1 via supplied adapter (sheet 16). CAN2 remains Delta only; keep two effective terminations per bus.

HOLD T01: Delta Gen 2 profile/bitrate, wheel calibration and traction-control failure behavior before enabling TC.

HOLD F02: combined R3 5V reference current, cockpit supply/return load and camera-kit variant/protection test.

03 / NEXUS R3 PIN ALLOCATION

ONE R3 with owner-selected HT-183200 flying-lead harness. Supplied controller leads need routing and free-end termination.

One wideband: onboard WBI1 / C29-C34. No external WB controller. Confirm sensor type/adapter; configure NSP.

A10/A11 are ground OUTPUTS: do not bond them directly to battery, engine or chassis. C26 is sensor return only.

DBW HBO1/2; HBO3 = ABS ignition; HBO4 = cockpit; HBO6 = PD16 wake. Separate protected feeds; verify each load.

HT-183200 supplies labeled flying leads, CAN plugs and wideband plug. Keep shielding; verify revision, reach and sensor adapters.

SPI3-6 remain spare; AVI11/C28 = HGT. A26 is NC: CARTEK U2 M6_IGN is the only source for S1 / R3 A13 (sheet 18).

04 / ENGINE POWER, INJECTION AND IGNITION

2011 L94 with LS3 cam/intake/throttle body | Donor connectors reused only after condition and cavity checks.

ELECTRONIC OUTPUT PROTECTION / NO DOWNSTREAM BANK FUSES

HCO1 -> all injector +12V branches; HCO2 -> all coil +12V branches.

Set fuse current / startup time / delay / retries in NSP.

Each channel senses the SUM of its connected loads.

All branch wires and contacts must tolerate the programmed trip curve.

25A is output capacity, NOT a selected protection setting.

HOLD P02: HCO1/2 must protect retained OEM wire/contact branches. Stock L94 coil and injector wiring is retained; see sheet 19.

PWR/DRV/SIG/PG/SG remain functional device IDs. Stock bank connector reference cavities are on sheet 19; inspect donor before wiring.

IGN and INJ channel number = cylinder number; firing order is an ECU configuration item, not a reason to reorder wires.

Do not merge HCO1/HCO2 through donor shared feeds. Coil low-reference grounding remains HOLD pending exact coil/R3 evidence.

07 / FUEL PUMPS, SENDERS AND COOLING

Two DW100 lift pumps -> surge tank -> DW200 feed pump | PROFORM 67049 kit: two 67048 brushless puller fans; 10 A each; independent on/off feeds.

LIFT PUMPS: SEPARATE PROTECTED CHANNELS

Lift 1 = R3 E4; lift 2 = PD16 B4. No F31/F32 inline fuses.

Set each trip curve for its own leads; verify cold/hot startup.

HT-010910: A ground / B +5V / C signal -> R3 AVI6 C15. Pull-up OFF; 0.5V = 0 PSI / 4.5V = 150 PSI.

Remote-mount HT-010910 with flexible fuel-rated hose; do not mount directly on rail/block. PROFORM pair: 20 A listed.

Lift pumps now have independent 25A-capable outputs. Select lower trip thresholds from measured load and lead/contact ratings.

Fuel permission requires engine run or timed prime AND valid PD16 communication/output status. Trip/CAN loss -> fuel/engine stop.

Fuel pumps: prime/run logic with engine-stall shutdown; service prime must be momentary/time-limited and master-stop subordinate.

Fans: PD16 B1/B2 supply independent switched +12V (100% when on). No motor-power PWM or external speed-control signal.

Fans: 11.43 x 10.75 x 3.25 in each; pair 22.86 x 10.75 in on 24 x 12 in core. Rear clearance confirmed; verify brackets/shroud.

12 / HARNESS CONVENTIONS AND OPEN ITEMS

This is a connected functional draft for review. It is not a fabrication release or an ECU configuration file.

HOW TO READ THIS SET

Identical Wxxx_NAME global labels are electrically connected across sheets.

Wxxx identifies an electrical net; branched nets require individual harness conductor IDs later.

Numeric R3/PD16/Delta/iC-7 pins are documented. Alphabetic functional IDs on TBD parts are not cavity numbers.

NC marks identify deliberately unused pins. A net ending TBD may have only one endpoint.

Reference J10/J11/J12 identifies R3 connectors A/C/E; J1 identifies its main power interface.

GROUNDING AND CONSTRUCTION

Battery/engine/chassis returns, R3 sensor return, R3 ground outputs and ABS sensor returns are distinct nets.

Do not bond R3 ground-output pins A10/A11 to chassis. Use A10 for shield drains at the ECU end only.

Reuse L94 donor plugs/pigtails after inspection; verify cavity identity and record color changes.

Wire sizes, insulation class, connector contacts, splice sizes and route lengths remain build-stage selections.

Electronic output protection must protect EVERY branch wire/contact; verify trip curves, cranking drop and startup.

RELEASE HOLDS

P01 CARTEK GT alone selected; verify engine-stop input, return isolation, cutoff timing, charging and starter driver.

E01 Crank/cam PNs identified; verify actual cavities, coil/injector/DBW hardware, 200mA reference and 500mA return budgets.

P02 Electronic trip curves vs weakest injector/coil/pump branches; F01 load/startup, fan settings and hot-idle test.

A01 ABS ground wording; A02 private/public CAN mapping; A03 C7 sensor signal/polarity compatibility.

N01 Gen 2 CAN profile/bitrate, TC loss behavior (sheet 15), PD16 setup; gear/HGT calibration and fuel-level input details.

CONFIGURATION AND COMMISSIONING

Load/program NSP separately: I/O mapping, output protection, sensor calibration, DBW faults and start permissives.

Check continuity/cavity orientation with electronics unplugged; then check supplies before connecting sensors.

Verify master cutoff with engine running, fuel stall shutdown, fan control and brake-light response.

Review source revisions and actual hardware before termination. No vehicle-powered test was performed.

NASA CCR is the requested safety reference; this drawing does not certify compliance.

Selected CAN traction-control path and spare I/O: sheet 15. Sources and pin/net schedule: README.md / Connections.tsv.

Verified manufacturer electrical limits are ceilings, not chosen fuse-trip settings. No fuse or wire purchase schedule is released.

17 / WOLFBOX G900 PRO CAMERA POWER

Keypad request -> R3 CAN control -> PD16 A7 HCO5. Complete power removal when commanded OFF; no parking supply.

SWITCHED-ONLY DESIGN / VERIFY KIT ON RECEIPT

Yellow BATT AND red ACC take the same camera-switched

12V feed from PD16 A7. Black returns to vehicle return.

This deliberately removes all camera power when OFF.

Retain the WolfBox converter and its supplied USB-C cable.

Use a kit matching the mirror version/input-current label.

No separate 12V or USB supply goes to the rear camera.

Select normal rear view; disable screen auto-off as needed.

CONTROL AND PROTECTION

Camera button toggles PD16 A7; proposed keypad position 10, initial OFF. R3 / PD16 / keypad must already be awake.

Use high-side ON/OFF, full supply when ON; do not PWM the converter supply. Verify orderly recording on power loss.

USB and rear-cable W labels denote complete vendor cable assemblies, NOT single wires or USB cavity numbers.

HOLD C01: confirm purchased kit labels/variant and switched-only operation with both BATT/ACC on A7 before installation.

3A/4A at 5V are mirror supply ratings, not measured continuous draw. At 12V / 90% efficiency: about 1.4A / 1.9A input.

Select PD16 trip/inrush/retry settings to protect actual kit leads; retain vendor inline protection if fitted. Do not use 8A blindly.

No raw-battery parking feed; camera stays downstream of master isolation. GPS and reverse lead omitted as the simple rear-view basis.

Insulate unused reverse lead. Verify boot time, rear view, CAN-loss OFF, cranking and master-stop behavior on actual hardware.

REV J / PHYSICAL CONDUCTOR VIEW 01 | 96 native component interfaces | 116 native nets | C179 / S85 / V2 / I11 / B9 / U1

Connection topology is proposed and geometry is illustrative. No measured cut lengths, fabrication release, installed NSP settings or powered verification claimed.